

Advanced (Habanero)

Q1. Factor the expression: $x^2 - 4$

- (A) $(x + 2)(x - 2)$
- (B) $x(x - 4)$
- (C) $x(x + 4)$
- (D) $2(x - 2)$
- (E) $2(x + 2)$

Q2. Simplify: $(2x)^2 - (3y)^2$

- (A) $4x^2 - 9y^2$
- (B) $4x + 9y$
- (C) $4x^2 + 9y^2$
- (D) $2x - 3y$
- (E) $2x + 3y$

Q3. Factor: $25a^2 - 9b^2$

- (A) $(5a - 3b)(5a + 3b)$
- (B) $(5a + 3b)(5a - 3b)$
- (C) $(25a - 9b)(a + b)$
- (D) $(25a + 9b)(a - b)$
- (E) $(5a - 9b)(5a + 3b)$

Q4. Solve for x : $x^2 - 16 = 0$

- (A) $x = pm16$
- (B) $x = pm4$
- (C) $x = 16$
- (D) $x = -16$
- (E) $x = 0$

Q5. Factor the expression: $x^2 - 9y^2$

- (A) $x(x - 3y)$
- (B) $x(x + 3y)$
- (C) $(x - 3y)(x + 3y)$
- (D) $(x + 3y)(x - 3y)$
- (E) $x(x - 9y)$

Q6. Simplify: $4x^2 - 9$

- (A) $2x - 3$
- (B) $(2x - 3)(2x + 3)$
- (C) $4x^2 + 9$
- (D) $2x + 3$
- (E) $4x - 9$

Q7. Factor: $x^2 - 1$

- (A) $x(x - 1)$
- (B) $(x - 1)(x + 1)$
- (C) $x(x + 1)$
- (D) $(x + 1)(x - 1)$
- (E) $x^2 + 1$

Q8. Simplify: $(x + y)^2 - (x - y)^2$

- (A) $4xy$
- (B) $4x^2 - 4y^2$
- (C) $4y^2 - 4x^2$
- (D) $2x + 2y$
- (E) $2x - 2y$

Open Ended Questions (FRQ)

Q9. Factor the expression: $x^4 - y^4$

Q10. Find the value of x in the equation $x^2 - 4x - 21 = 0$ using difference of squares. Show your steps.

Chef's Tips

- Remind students of the formula: $a^2 - b^2 = (a + b)(a - b)$
- Make sure students check their work
- Encourage students to use their calculators to verify their answers